DESIGN OF MIL–STD–461 COMPLIANT EMI FILTERS FOR FLYBACK DC–DC CONVERTER

A PROJECT REPORT

Submitted in Partial Fulfilment for the degree of

Bachelor of Technology in

ELECTRONICS & COMMUNICATION ENGINEERING

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He has successfully completed the project. During this period, he has acquired good knowledge in various converter topologies, EMI filter design, SPICE simulations and frequency domain analysis. During the period, his character and conduct were found good. I wish him a bright future and a successful career.

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ABSTRACT

This report looks into the important issue of Electromagnetic Interference (EMI) caused by a switch-mode flyback DC–DC converter. It offers a clear method to achieve efficient voltage conversion while reducing noise emissions through the design and integration of an effective EMI filter. Flyback converters are commonly used in low-to-medium power applications due to their simplicity, galvanic isolation, and ability to provide multiple outputs. However, their high-frequency switching and discontinuous current mode operation produce significant conducted and radiated noise. This noise can harm system performance and lead to failures in meeting strict regulatory standards.

To tackle this problem, a two-stage EMI filter is added at the input. This filter effectively reduces common-mode and differential-mode noise without compromising the converter's performance. The design process includes identifying the main noise sources, modelling the converter's impedance, and choosing filter components to achieve the needed noise reduction as outlined by military standards like MIL–STD 461G. The report also details the simulation of the filter's performance and its effects on the converter's overall efficiency and stability. It addresses potential issues such as power factor degradation and resonance with the converter's input impedance. Lastly, the report presents experimental results from a prototype build, which confirm the filter's ability to reduce EMI to acceptable levels while keeping the converter's operational integrity intact.

Chapter – 1 : Introduction

In today's fast-changing electronics industry, the need for efficient, compact, and reliable power conversion systems has grown significantly. DC-DC converters are crucial in power management circuits. They convert a specific level of DC voltage into another level needed by various electronic devices. These converters are commonly used in industrial, automotive, and consumer electronics where stable and efficient voltage regulation is important. Among the different DC-DC converter designs, the flyback converter is notable for its simplicity, low component count, and ability to provide electrical isolation between the input and output. This isolation improves safety and protects low-voltage circuits from high-voltage disturbances. It makes the flyback converter suitable for low- to medium-power applications such as battery chargers, adapters, and auxiliary power supplies.

However, one major challenge with switching converters like the flyback converter is the generation of Electromagnetic Interference (EMI). The high-frequency switching of power semiconductor devices causes rapid changes in voltage and current, leading to both conducted and radiated EMI. This noise can interfere with nearby electronic devices or even impact the converter's performance. To comply with regulatory standards on Electromagnetic Compatibility (EMC) and to ensure reliable operation, it is crucial to include effective EMI reduction techniques in the converter design.

An EMI filter is introduced at the input stage of the converter to reduce unwanted noise signals before they travel through the power lines. The filter is designed to suppress both common-mode and differential-mode noise without affecting the power transfer or the converter's efficiency. This results in cleaner power delivery and compliance with international EMI standards.

For military systems, ensuring Electromagnetic Compatibility (EMC) is not merely a design goal but a mandatory requirement for system reliability and interoperability. All electronic subsystems must comply with stringent standards, with MIL-STD 461G being the governing document for the control of EMI characteristics. The conducted emissions (CE) requirements of this standard, particularly CE102 (10 kHz to 10 MHz), set strict limits on the noise that equipment is allowed to inject back onto its power lines. A standard, unfiltered flyback converter will fail these limits by a significant margin.

This report examines the design, modelling, and implementation of a Flyback DC-DC converter integrated with an EMI filter. The study includes theoretical analysis, simulations using relevant design software, component selection based on performance needs, and experimental testing of the prototype. The performance of the system is assessed in terms of output voltage regulation, efficiency, and noise reduction. The goal of this work is to create a compact and efficient power converter that provides a reliable DC output while ensuring electromagnetic compatibility for modern electronic applications.

Chapter – 2 : Project Fundamentals

2.1 Converters

A converter is a key power electronic device that changes electrical energy from one form to another to meet the needs of various electronic systems. Generally, converters regulate voltage, current, or frequency to provide a steady and efficient power supply to loads like microcontrollers, communication systems, industrial machines, and consumer electronics. By using semiconductor switching devices like MOSFETs, IGBTs, or diodes, converters achieve high efficiency and a compact design compared to traditional linear power supplies.

Converters can be divided into four main categories based on the type of input and output signals they can handle :

I. **AC to DC Converters (Rectifiers) :** These converters change Alternating Current (AC) input into Direct Current (DC) output. They are commonly found in power supplies for electronic devices, where AC mains voltage is rectified and filtered to produce a steady DC voltage.

II. **DC to AC Converters (Inverters) :** These converters do the opposite of rectifiers by converting DC input into AC output. They are widely used in renewable energy systems such as solar inverters, uninterruptible power supplies (UPS), and motor drive applications.

III. **DC to DC Converters :** These converters change a DC voltage from one level to another. They are very efficient and compact, making them perfect for battery – powered systems and embedded electronics.

IV. **AC to AC Converters :** These converters adjust an AC input to another AC output with a different voltage, frequency, or phase. Examples include cycloconverters and AC voltage controllers, which are often used in variable-speed motor drives and voltage regulation systems.

2.1.1 DC – DC Converters

A DC–DC converter is a type of power electronic circuit designed to convert a direct current (DC) voltage from one level to another. It plays a crucial role in applications where different subsystems require distinct operating voltage levels.

2.1.1.1 Basic Principle of Operation

The fundamental working principle of a DC–DC converter is based on energy storage and transfer through inductive or capacitive elements. Semiconductor switches such as MOSFETs or IGBTs are rapidly turned ON and OFF at high frequencies, controlling the average voltage and current supplied to the load. During each switching cycle:

- Energy is stored in an inductor or transformer when the switch is ON.
- The stored energy is transferred to the output when the switch is OFF.

By adjusting the duty cycle (the ratio of ON time to total switching period), the converter can regulate the output voltage according to the required level

2.1.1.2 Classifications of DC – DC Converters

2.1.1.2.1 Non-Isolated DC–DC Converters

- **Buck Converter (Step-Down Converter) :** The buck converter reduces the input voltage to a lower output voltage. Output voltage relation :

$$V_o = D \times V_{in}$$

[Equation – 1]

where D is the duty cycle.

- **Boost Converter (Step-Up Converter) :** The boost converter increases the input voltage to a higher level. Output voltage relation :

$$V_o = \frac{Vin}{1 - D}$$

[Equation – 2]

- **Buck-Boost Converter :** This converter can either increase or decrease the input voltage depending on the duty cycle. However, the output polarity is inverted with respect to the input.

  Output voltage relation :

$$V_o = -\frac{D}{1 - D} \times V_{in}$$

[Equation – 3]

- **SEPIC (Single-Ended Primary Inductor Converter) and Cuk Converter :** These are advanced non-isolated topologies capable of both step-up and step-down conversion while maintaining non-inverted or inverted outputs, respectively. They provide better control over current and voltage ripples.

2.1.1.2.2 Isolated DC–DC Converters

- **Flyback Converter :** The flyback converter is a modified buck–boost topology that uses a transformer for isolation and voltage scaling. Energy is stored in the transformer's magnetic core during the ON period and released to the output during the OFF period. It is simple, cost effective, and ideal for low to medium power levels (typically up to 100 W).

- **Forward Converter :** In this topology, energy is transferred directly from the input to the output through the transformer during the switch ON time. It offers higher efficiency and lower voltage stress on components compared to the flyback converter.

- **Push-Pull, Half-Bridge, and Full-Bridge Converters :** These are advanced isolated topologies designed for medium to high-power applications. They use multiple switches and transformer windings to achieve bidirectional power flow and reduced ripple, making them suitable for high-performance systems.

2.1.1.3 Why Flyback Converter is Used here

The Flyback Converter is selected for this project because it combines simplicity, cost-effectiveness, and electrical isolation. This makes it suitable for low- to medium-power DC–DC conversion applications. Its unique operation allows it to step up and step-down voltage while providing electrical isolation between the input and output circuits through a transformer. The following points outline the main reasons for choosing the flyback topology for this design :

- **Electrical Isolation and Safety :** A key reason for using a flyback converter is its ability to provide electrical isolation between the input and output. The isolation transformer prevents a direct electrical connection between high-voltage and low-voltage sides, ensuring user safety and protecting sensitive components. This feature is especially important in applications where the output must be electrically isolated from the input source, such as in chargers, adapters, and industrial control systems.

- **Simplicity of Design :** The flyback converter has a simple circuit design, needing only one switching transistor, a transformer, and a few passive components. Unlike other isolated designs like forward, push-pull, or bridge converters, the flyback converter does not require extra output inductors or complex drive circuitry. This simplicity helps reduce the overall cost, size, and complexity of the design, making it a practical choice for compact power supply units.

- **Cost – Effectiveness :** Due to its low component count and straightforward construction, the flyback converter is more affordable compared to other isolated converters. It removes the need for multiple switches or magnetic components, lowering both manufacturing and maintenance costs. This benefit is especially important in consumer-grade and low-power industrial applications where balancing cost and efficiency is crucial.

- **Versatility and Wide Input Range :** The flyback converter can operate over a broad input voltage range, making it suitable for applications that require universal input (for example, 85–265 V AC after rectification). Its ability to both step–up and step–down voltage allows it to adapt to different supply conditions without major circuit changes. This versatility makes the converter useful in various environments.

- **Energy Storage and Regulation :** The flyback converter inherently stores energy in the magnetic field of the transformer during the switch's ON period and releases it to the output during the OFF period. This dual mechanism for energy storage and transfer provides excellent voltage regulation and load adaptability. With proper feedback control, the converter can keep a stable DC output even under changing load conditions.

- **EMI Filtering and Noise Control :** Although switching converters create high-frequency noise, the flyback converter's isolated transformer helps reduce common-mode noise between input and output. When used with an EMI filter, it effectively suppresses conducted and radiated emissions, ensuring compliance with Electromagnetic Compatibility (EMC) standards. This makes the flyback converter ideal for environments where low noise levels are essential.

- **Suitability for Low and Medium Power Applications :** The flyback converter works best at power levels typically below 100 W, which matches many practical uses, such as chargers, adapters, and auxiliary power supplies. Its high efficiency, compact design, and isolation features make it an optimal choice for these power levels.

2.2 Modes of Current Operation in Flyback Converter

The operation of a Flyback Converter largely depends on how the current flows through the transformer windings and the energy storage process during each switching cycle. Based on the behaviour of the magnetizing current in the transformer, the converter can operate in two distinct modes:

- Continuous Conduction Mode (CCM)
- Discontinuous Conduction Mode (DCM)
- Boundary Conduction Mode (BCM/CrCM)
- Quasi-Resonant Mode (QRM/Valley Switching)
- Hysteretic Conduction Mode (HCM)

2.2.1 Continuous Conduction Mode (CCM)

In Continuous Conduction Mode, the magnetizing current in the transformer never falls to zero during the switching cycle. This means that some amount of energy remains stored in the transformer's magnetic core at all times.

Operation :

1. **T_{on} (MOSFET On) :** Primary current (I_p) ramps up from zero. Simultaneously, the secondary current (I_s) is forced to zero (as the diode becomes reverse-biased), but this transition is not instantaneous.

2. **T_{off} (MOSFET Off) :** Secondary current (I_s) ramps down from its peak, but does not reach zero before the T_{on} cycle begins again.

Characteristics :

In CCM operation, the current waveform through the transformer is smoother, and the peak current values are lower compared to Discontinuous Conduction Mode (DCM). This reduces conduction losses and usually results in higher efficiency at moderate to heavy load conditions. The continuous energy transfer also results in smaller output voltage ripple, which improves voltage

stability. However, running in CCM requires a larger magnetic core to handle the continuous current, which slightly increases the size and cost of the converter.

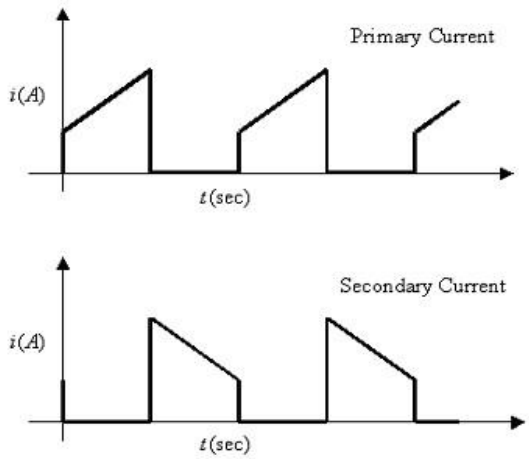

Figure 1 – Primary and Secondary Current in CCM

Advantages :

1. Reduced current stress on the switch and components.
2. Improved efficiency and lower EMI due to smoother current waveform.

Disadvantages :

1. More complex control loop design due to right-half-plane zero (RHPZ) in small-signal response.
2. Larger magnetic components required, increasing size and cost.

2.2.2 Discontinuous Conduction Mode (DCM)

In DCM, the energy stored in the transformer's core during the "on" state of the MOSFET is completely transferred to the output and depleted before the next switching cycle begins. The secondary current (I_s) drops to zero and remains there for a portion of the switching period.

Operation :

1. **T_{on} (MOSFET On) :** Primary current I_p ramps up, storing energy. I_s is zero.

2. **T_{off} (MOSFET Off) :** Secondary current I_s starts at its peak and ramps down as energy is transferred to the output.

3. **T_{idle} (Dead Time) :** Is reaches zero. Because the MOSFET is still off, no current flows in either winding. The converter "idles" until the next T_{on} cycle.

Characteristics :

In Discontinuous Conduction Mode, the primary current in the transformer drops to zero before the next switching cycle begins. This happens because all the energy stored during the switch ON period is fully transferred to the output during the OFF period. This leaves a short gap of zero current before the next cycle starts. Consequently, the peak current in DCM is higher than in Continuous Conduction Mode (CCM), which increases stress on the switching components. However, this mode works well for low to medium power applications and for situations where the converter runs under light loads.

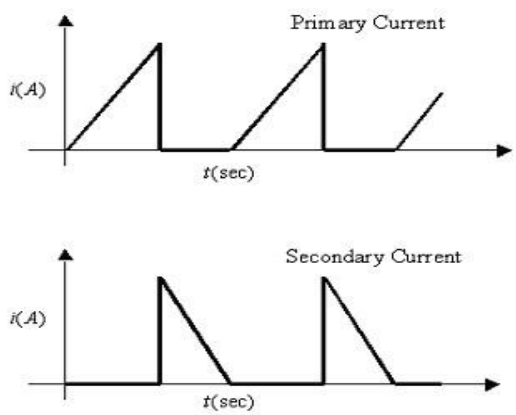

Figure 2 – Primary and Secondary Current in DCM

Advantages :

1. Simpler control design and faster transient response.
2. Reduced magnetic core size due to full energy transfer per cycle.
3. Lower core losses at light loads.

Disadvantages :

1. Higher peak currents lead to increased switching and conduction losses.

2. Greater EMI due to sharp current transitions.
3. Higher output voltage ripple compared to CCM.

2.2.3 Boundary Conduction Mode (BCM) or Critical Conduction Mode (CrCM)

Operation :

In this mode the inductor or transformer current begins at zero, rises during the switch on time, and returns to zero exactly at the end of the off-time. The controller immediately turns the switch on when current reaches zero. The converter therefore operates precisely at the boundary between continuous and discontinuous conduction.

Characteristics :

The current waveform touches zero every cycle but does not stay at zero for any significant duration. Switching frequency varies with load because the timing depends on when the current reaches zero. Peak current is higher than in CCM but lower than in DCM. Zero-current switching occurs naturally, reducing switching losses.

Advantages :

1. Lower switching losses because the switch turns on at zero current.
2. Improved efficiency compared to CCM at light to medium loads.
3. Simplified control compared to DCM because current detection is straightforward.
4. Good compromise between low ripple and low losses.

Disadvantages :

1. Variable switching frequency complicates EMI filter design.
2. Higher peak current stresses components more than CCM.
3. Requires zero-current detection circuitry.
4. Not ideal for heavy loads.

2.2.4 Quasi-Resonant Mode (QRM/Valley Switching)

Operation :

The switch is turned on at the minimum point of the drain-source voltage waveform, using the natural resonant ringing between the inductor and parasitic capacitances. The controller waits for a valley to appear and switches at that instant, which reduces the energy lost during switching.

Characteristics :

Switching frequency varies significantly with load because the valley positions change with operating conditions. The voltage waveform across the switch oscillates after each cycle and the control system aligns the turn-on moment with the lowest voltage. Resonant behaviour produces softer transitions and reduced switching stress.

Advantages :

1. Very low switching losses due to turning on at minimum voltage.
2. Reduced electromagnetic interference because transitions are softer.
3. Improved overall efficiency compared to fixed-frequency CCM.

Disadvantages :

1. Highly variable switching frequency complicates magnetic design and EMI compliance.
2. Requires more complex control circuits for valley detection.
3. Component stresses can be unpredictable at very light loads.

2.2.5 Hysteretic Conduction Mode (HCM)

Operation :

The controller regulates current by allowing it to swing between two defined limits. When the current reaches the upper threshold, the switch turns off. When it falls to the lower threshold, the switch turns on again. This creates a direct current-controlled behaviour independent of a fixed PWM timing circuit.

Characteristics :

The switching frequency naturally varies with load because the time required to move between the current thresholds depends on operating conditions. The current waveform forms a sawtooth shape bounded by the hysteresis band. Response to load changes is very fast because the controller reacts immediately when a threshold is reached.

Advantages :

1. Excellent transient performance because of instantaneous control action.
2. Simple control implementation with no need for a traditional PWM loop.
3. Good efficiency at light loads.

Disadvantages :

1. Variable frequency makes EMI filtering more difficult.
2. Ripple may be higher due to large current swings.
3. Less suitable for applications requiring constant switching frequency.

2.3 PWM Current mode Controller (LT1241)

The LT1241 is a high-speed, fixed-frequency, housed in an 8-pin package, current-mode PWM (Pulse Width Modulation) controller made by Analog Devices, which was formerly known as Linear Technology. It is commonly used in DC-DC converters and off-line switch-mode power supplies, including flyback converters. The datasheet offers complete technical details needed for designing and implementing circuits with this IC, such as its features, pin layout, electrical specs, and application guidelines.

The LT1241 is a better version of the standard UC1842 series, maintaining pin compatibility while providing faster response times and lower quiescent current. It includes several important components: a bandgap voltage reference, an error amplifier, a current-sense comparator, an oscillator, and a high-current totem-pole output stage for driving power MOSFETs directly. The IC is optimized for high-frequency operation up to 500 kHz, making it ideal for small, high-efficiency converter designs. It also has a low start-up current of less than 250 μA, which helps reduce power loss during initial operation.

2.3.1 Pin Configuration

Pin	Name	Function
1	COMP	Compensation : Output of the error amplifier. Used for external loop compensation.
2	FB	Feedback : Inverting input to the error amplifier. Receives a portion of the converter's output voltage.
3	I_{SENSE}	Current Sense : Input to the current sense comparator. Monitors the voltage from the current sense resistor.
4	R_T/C_T	Timing : An external resistor (R_T) and capacitor (C_T) are connected to this pin to set the oscillator frequency.
5	GND	Ground : Common ground for the device.
6	OUTPUT	Output : High-current totem pole driver output for the external power MOSFET
7	V_{CC}	Power Supply : The positive supply voltage for the IC.
8	V_{REF}	Reference Output : Provides a 5V trimmed reference voltage.

Table 1 – Pin Configuration of LT1241

2.3.2 Principle Features and Protection Mechanisms

1. **Leading Edge Blanking (LEB) :** This feature is vital for current-mode controllers. The IC's internal logic ignores the output from the current sense comparator for a short time, usually 100ns, right after the output switch turns on. This helps prevent the comparator from triggering too early due to the "leading edge" current spike from transformer interwinding capacitance and output diode reverse recovery. This feature often removes the need for an external RC filter on the current sense pin, which improves the controller's response time.

2. **Undervoltage Lockout (UVLO) :** The device has a UVLO circuit with hysteresis to ensure a reliable startup and shutdown process. The controller will not start until the V_{CC} supply (Pin 7) reaches a set threshold, typically 9.6V for LT1241. During UVLO, the startup current is very low, about 170μA typically and a maximum of 250μA, making it ideal for off-line supplies.

3. **Active Pull-Down :** When in a UVLO state, both V_{REF} (Pin 8) and OUTPUT (Pin 6) pins are actively pulled low. The pull-down on the output pin keeps the external power MOSFET completely off, preventing it from turning on by accident during power-up or power-down.

4. **Internal 5V Reference :** A trimmed bandgap reference provides a stable 5V output at the V_{REF} pin (Pin 8). This reference powers the oscillator's R_T network and generates the internal 2.5V reference for the error amplifier.

5. **Output Protection :** The output stage includes an 18V clamp. This protects the gate of the external MOSFET from high voltage, especially during transient conditions.

2.3.3 Block Diagram of the Chip

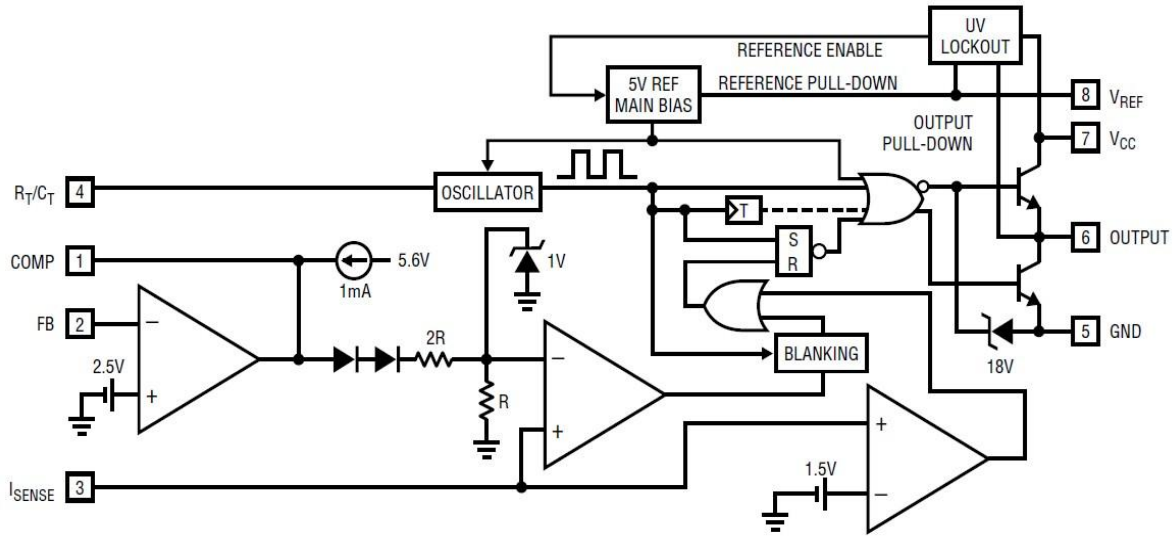

Figure 3 – Block Diagram of IC LT1241

2.3.4 Core Functional Blocks

2.3.4.1 Oscillator

The device uses a fixed-frequency oscillator, with the operating frequency and maximum duty cycle set externally by a resistor (R_T) and a capacitor (C_T) connected to the R_T/C_T pin. The oscillator provides the internal clock signal that initiates each PWM cycle.

2.3.4.2 Error Amplifier :

A high-gain error amplifier is used to regulate the converter's output voltage. It compares a feedback signal from the converter's output, fed into the inverting input, against a stable internal

2.5V reference. The output of this amplifier (COMP, Pin 1) sets the voltage threshold for the current-sense comparator.

2.3.4.3 Current-Mode Control and PWM Latch :

This is the core of the current-mode operation. The voltage at the I_{SENSE} pin, which is typically generated by a sense resistor in series with the power MOSFET, represents the instantaneous switch current. This voltage is fed into the current sense comparator.

- At the start of each oscillator cycle, the PWM latch turns on the output driver.
- The switch current (and thus the I_{SENSE} voltage) ramps up.
- When the I_{SENSE} voltage exceeds the threshold set by the Error Amplifier's output, the comparator trips the latch, turning the output driver off for the remainder of the cycle.
- This mechanism provides cycle-by-cycle current limiting and simplifies the loop compensation.

2.4 The Electromagnetic Compatibility (EMC) Tests : MIL–STD–461G

2.4.1 Standard Overview and Applicability

The electromagnetic compatibility (EMC) requirements for the project are defined by MILSTD-461G, a U.S. military standard that establishes the required control of electromagnetic interference (EMI) characteristics for electronic equipment and subsystems. Adherence to this standard ensures that equipment can operate reliably in its intended electromagnetic environment without causing or suffering degradation due to EMI. In the Table for the tests (Figure - 4), the legend uses three critical abbreviations to specify the degree of necessity for each test method :

- **A [Applicable] :** This designation means the test is required and must be performed on the Equipment Under Test (EUT) to demonstrate compliance with the standard. If a test is marked 'A' for the EUT's platform, there is no exception, and the test must be executed following the general requirements of that specific test method.

- **L [Limited] :** This designation means the test is required but subject to modifications or waivers as specified in the individual sections of the MIL-STD-461G standard itself. Often, the limits or frequency ranges are relaxed, restricted, or tailored based on the specific operational characteristics of the platform (e.g., lower frequency range limits or different test voltages).

- **S [Procuring activity must specify in procurement documentation]** : This designation indicates that the test is conditionally required. It is the responsibility of the system buyer (the procuring activity, e.g., the specific government agency) to determine if the test is necessary based on platform specific threats or environment. If the procuring agency fails to specify, the test is typically not required by default.

Equipment and Subsystems Installed In, On, or Launched From the Following Platforms or Installations	Requirement Applicability																		
	CE101	CE102	CE106	CS101	CS103	CS104	CS105	CS109	CS114	CS115	CS116	CS117	CS118	RE101	RE102	RE103	RS101	RS103	RS105
Surface Ships	A	A	L	A	S	L	S	L	A	S	A	L	S	A	A	L	L	A	L
Submarines	A	A	L	A	S	L	S	L	A	S	L	S	S	A	A	L	L	A	L
Aircraft, Army, Including Flight Line	A	A	L	A	S	S	S		A	A	A	L	A	A	A	L	A	A	L
Aircraft, Navy	L	A	L	A	S	S	S		A	A	A	L	A	L	A	L	L	A	L
Aircraft, Air Force		A	L	A	S	S	S		A	A	A	L	A		A	L		A	
Space Systems, Including Launch Vehicles		A	L	A	S	S	S		A	A	A	L			A	L		A	
Ground, Army		A	L	A	S	S	S		A	A	A	S	A		A	L	L	A	
Ground, Navy		A	L	A	S	S	S		A	A	A	S	A		A	L	L	A	L
Ground, Air Force		A	L	A	S	S	S		A	A	A		A		A	L		A	

Legend:
A: Applicable
L: Limited as specified in the individual sections of this standard.
S: Procuring activity must specify in procurement documentation.

Table 2 – A Table of type of subsystem and the required tests for that subsystem

As the project is designed for Space Systems, including Launch Vehicles, the full applicability matrix within MIL-STD-461G is consulted to determine the required test methods. The standard lists numerous Conducted Emissions (CE), Conducted Susceptibility (CS), Radiated Emissions (RE) and Radiated Susceptibility (RS) tests. However, many tests are discounted due to the project's specific configuration and simulation focus. For example, tests related to antenna ports (such as CS103, CS104, CS105, and CE106) and cable-coupled interference (CS114 through CS117) are deemed nonapplicable or insignificant. Furthermore, radiated tests (RE and RS) are not considered critical for the scope of this particular simulation. Therefore, for this project, only the following two tests are considered applicable and necessary for compliance: CE102 and CS101.

2.4.2 Applicable Test Methods

2.4.2.1 CE102: Conducted Emissions, Power Leads (10 kHz to 10 MHz)

The CE102 test measures unwanted electromagnetic energy conducted from the Equipment Under Test (EUT) back onto its power leads. This test is essential to ensure that the equipment does not inject excessive noise onto the platform's power distribution system, which could affect other connected subsystems. The CE 102 test is conducted by taking the FFT of the LISN output in the main circuit. Once compliance with the given limits, then the CS 101 (will be discussed later) test will be conducted.

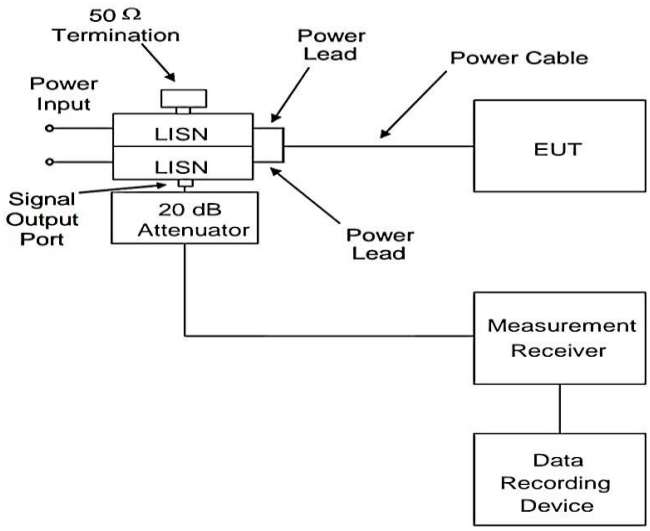

Figure 4 – The Block diagram of Measurement Setup for the CE 102 Test

The limits for the CE102 test are typically defined by a "Basic Curve" limit level in dBμV. The limit level is highest at lower frequencies (e.g., 94 dBμV at 10 kHz) and rolls off logarithmically to a constant level (e.g., 60 dBμV) from 700 kHz to 10 MHz. Adjustments to this basic curve, known as Limit Relaxation, are applied based on the EUT's nominal source voltage (AC or DC).

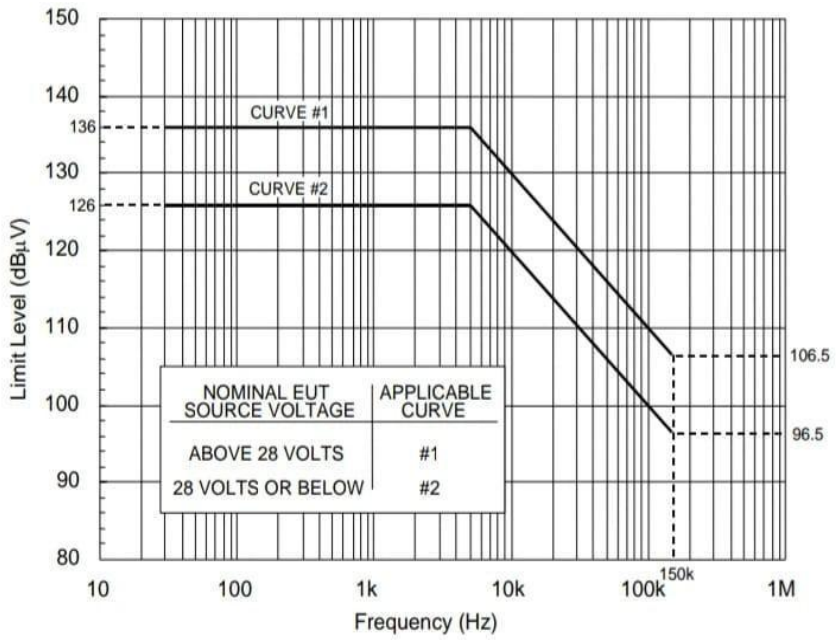

Figure 5 – The Basic Reference Curve for the CE 102 Test

2.4.2.2 CS101: Conducted Susceptibility, Power Leads (30 Hz to 150 kHz)

The CS101 test assesses the equipment's ability to operate without degradation when subjected to low-frequency AC ripple and transient voltage spikes on its primary power input. This ensures the equipment is resilient against power line noise originating from other components or system transients.

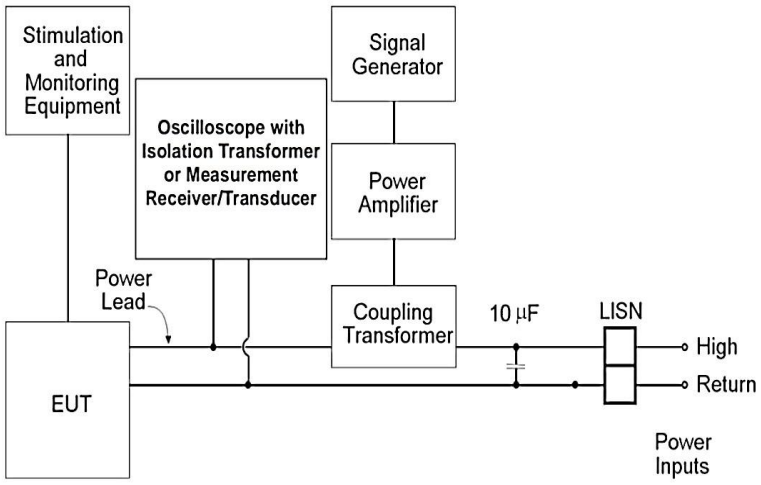

Figure 6 – Block Diagram of Measurement setup for CS101 test

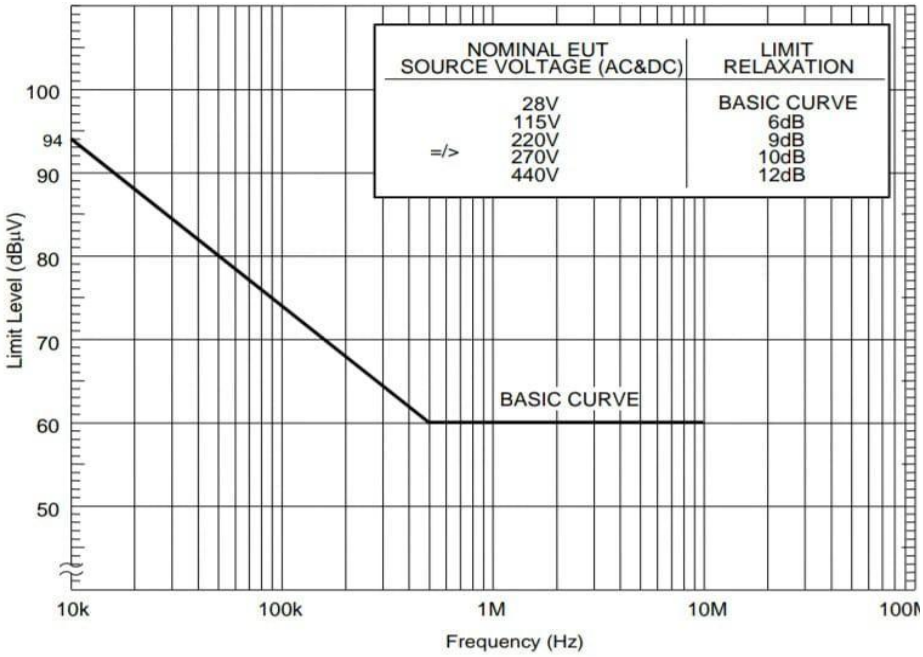

Figure 7 – The Basic Reference Curve for the CS 101 Test

The limit for CS101 is an injected voltage level defined by two curves based on the EUT's nominal source voltage :

- **Curve 1 :** Applicable for source voltages above 28 Volts. This curve sets a higher susceptibility limit, typically reaching up to 136 dBµV in the 30 Hz to 1 kHz range and decreased drastically from 5 kHz to 150 kHz upto 106.5 dBµV. In this project, this curve is used as the majority voltage in the range is above 28V.

- **Curve 2 :** Applicable for source voltages 28 Volts or below. This curve sets a lower susceptibility limit, typically reaching up to 126 dBµV in the 30 Hz to 1 kHz range and similarly like previous it also decreased drastically from 5 kHz to 150 kHz upto 96.5 dBµV.

2.4.2.2.1 CS 101 Calibration Circuit

2.4.2.2.1.1 Definition

A Calibration Circuit in the context of electromagnetic compatibility (EMC) testing, such as MIL-STD-461G CS101, is a non-destructive test setup used to establish and verify the performance of the injection hardware *before* it is connected to the Equipment Under Test (EUT).

The fundamental purpose of the calibration circuit is to ensure that the test equipment—specifically the signal generator, power amplifier, and injection device (like a transformer or coupling

network)—can produce the exact specified voltage or current levels across the required frequency range, into a standardized load.

For CS101, the calibration circuit, as shown in the diagram, typically consists of a coupling transformer (L1/L2) and a 0.5 Ω resistive load (R1). This 0.5 Ω resistor simulates the impedance of the Equipment Under Test (EUT) power lines as defined by the standard.

2.4.2.2.1.2 Circuit Diagram

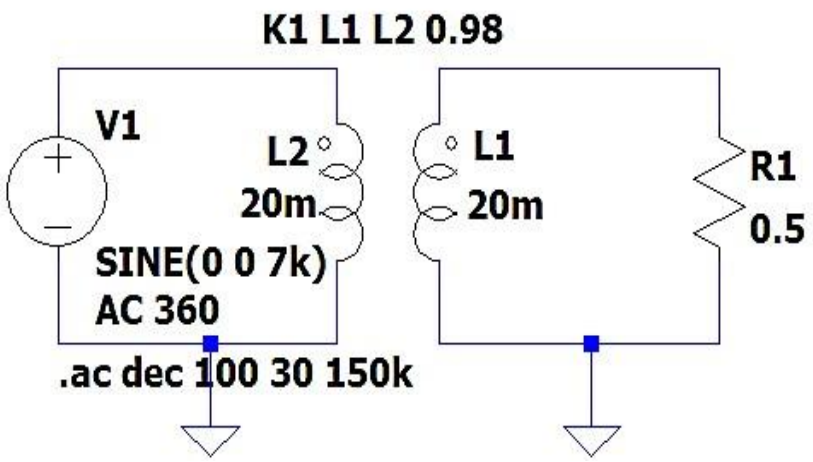

Figure 8 – The Circuit Diagram for the Calibration Circuit for CS 101 Test

2.4.2.2.1.3 Function of the CS101 Calibration Circuit

The calibration circuit is used to characterize the required input voltage (V1) necessary to produce the precise AC voltage noise on the secondary side (across R1) that meets the MIL-STD461G limits.

1. **Verification :** The signal generator is swept across the test frequency range (e.g., 30 Hz to 150 kHz).

2. **Measurement :** The voltage across the 0.5 Ω load (R1) is measured and compared against the limits (Curve 1 or Curve 2).

3. **Adjustment :** The signal generator's output (V1) is adjusted until the measured voltage across the load matches the required MIL-STD-461G susceptibility limit at every frequency point.

4. **Documentation :** The required input voltage values are documented or programmed into the test system.

2.4.2.2.2 CS101 Susceptibility Limits

The MIL-STD-461G CS101 test requires injecting a specific AC voltage noise level that varies with frequency, depending on the EUT's nominal source voltage. These limits ensure the equipment can withstand typical AC ripple and transient voltage spikes encountered in the operational environment.

Injection Frequency (HZ)	AC Amplitude (V)	Noise Level (dBV)	Noise Level (dBμV)
30	11.8	16	136
50	12	16	136
100	13.2	16	136
300	22.5	16	136
500	34	16	136
700	46.5	16	136
1k	65.5	16	136
1.36k	89	16	136
5k	325	16	136
7k	360	14	134
10k	320	10	130
30k	308.7	0	120

50k	320	-4	116
70k	318	-7	113
100k	320	-10	110
120k	305	-12	108
150k	302	-14	106

Table – 3 : The required injected voltage limits from the MIL-STD-461G curves

Chapter – 3 : The Project Standards

3.1 Project Objectives

1. Input Voltage Range (V_{in}) :

The converter must operate reliably across a wide DC input voltage range from 20 V to 40 V. This range defines the necessary component selection (e.g., MOSFET voltage rating) and the design requirements for the controller to handle wide duty-cycle variations.

2. Minimum Output Power ($P_{out, min}$) :

The system must deliver a stable minimum output power of 20W, corresponding to a required output of 5V at 4A. This confirms the efficiency and thermal management goals at minimum load conditions.

3. Switching Frequency (f_{sw}) :

The MOSFET switching frequency is set to 400 kHz. This frequency dictates the design of the transformer, the reactive filter components (L and C), and is a primary source of the high frequency noise targeted by the CE102 conducted emissions test.

4. EMC Compliance Goals (Conducted Emissions and susceptibility) :

The converter must demonstrate full compliance with the MIL-STD-461G CE102 limits, ensuring internal switching noise is suppressed by the EMI filter and does not exceed the allowed dBμV limits on the power leads.

5. EMC Compliance Goal (Conducted Susceptibility) :

The converter must withstand the injected AC ripple and transients specified by the MIL-STD461G CS101 limits without suffering performance degradation.

6. Susceptibility Performance Criterion :

The critical measure of success for the CS101 test is that the output voltage fluctuation must be strictly limited to 70 millivolts (mV) across the entire frequency sweep.

3.2 Block Diagram of the Main Circuit

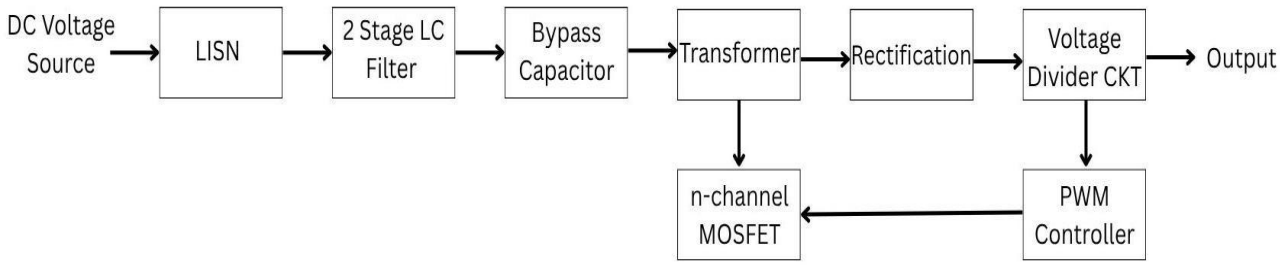

Figure 9 – The Block diagram of Process to be done

Chapter – 4 : Project Work Flow

4.1 The Open Loop Flyback Converter

The circuit (Figure 4) here represents an open-loop Flyback DC-DC converter. It contains a DC input and a capacitor of a very low value (0.1 μF) at the primary side, which mainly serves to filter high-frequency switching noise rather than provide large energy storage. A flyback converter belongs to the family of isolated switched-mode power supplies (SMPS) and operates on the principle of storing energy in the magnetic field of a coupled inductor (transformer) during one part of the switching cycle and releasing it during another.

4.1.1 Circuit Diagram

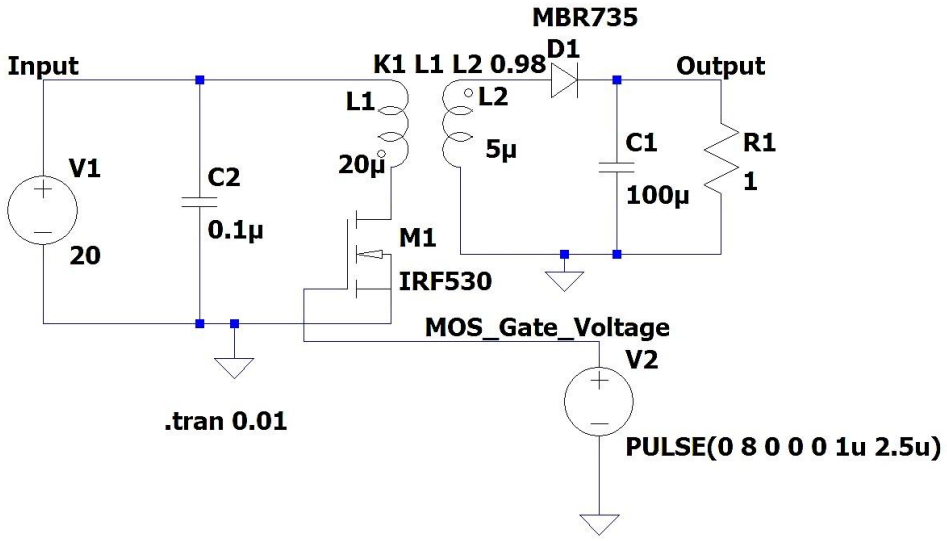

Figure 10 – Circuit Diagram of Open Loop Flyback Converter

4.1.2 Operational Principle

The converter's operation is defined by two distinct modes based on the state of the switch M1. The circuit's operation is cyclic, driven by the switching action of the MOSFET (M1) which is controlled by a Pulse Width Modulation (PWM) signal (V2). This operation is generally described in two distinct phases corresponding to the state of the switch. The operation can be divided into two distinct phases.

Phase 1 : The "ON" State (Energy Storage)

The first phase, known as the "ON" state or Energy Storage phase, begins when the control voltage V2 goes high to 8V, turning on the MOSFET M1 (IRF530). This action creates a closed loop on the primary side, allowing current to flow from the DC Input source V1 (20V) through the primary winding of the coupled inductor L1 (20μH) and the MOSFET to ground. During this interval, energy is effectively stored in the magnetic field of the transformer core, which has a coupling coefficient (K1) of 0.98, causing the current in the primary winding to rise linearly. Simultaneously, due to the winding polarity indicated by the dot convention, the voltage induced across the secondary winding L2 (5μH) is negative at the anode of the Schottky diode D1 (MBR735). Consequently, D1 remains reverse-biased and blocks current flow, meaning that the load R1 (1Ω) is supplied entirely by the energy previously stored in the output capacitor C1 (100μF) during this period.

Phase 2 : The "OFF" State (Energy Transfer)

The second phase, the "OFF" state or Energy Transfer phase, occurs when the control voltage V2 drops to 0V, turning the MOSFET M1 off and interrupting the primary current. The sudden collapse of the magnetic field induces a voltage reversal across the transformer windings to maintain flux conservation. This polarity flip makes the dot end of the secondary winding L2 positive, which forward-biases the diode D1 and allows it to conduct. The energy stored in the magnetic field during the ON phase is then transferred to the secondary side, where current flows through D1 to charge the output capacitor C1 and supply power to the load R1. While not explicitly shown in the simplified schematic, a snubber circuit is typically required across the primary winding to protect the MOSFET from voltage spikes caused by leakage inductance during this transition.

4.1.3 Component Analysis

- **Input Voltage, V_1 :** 20 - 40V DC input sources.
- **Capacitor, C_2 :** A 0.1μF primary-side capacitor. This value is too small for bulk energy storage (which handles the pulsed input current) and therefore serves as a high-frequency decoupling capacitor, designed to filter high-frequency switching noise.
- **MOSFET, M1 (Model – IRF530) :** An N-channel MOSFET acting as the main power switch.
- **MOSFET Gate Voltage, V_2 :** A PULSE voltage source that provides the gate drive signal for M1.
- **Coupled Inductor (L_1/L_2) :** The flyback transformer.

- **D1 (MBR735) :** A Schottky diode that acts as the output rectifier. It is fast-switching and has a low forward voltage drop, which is essential for high-frequency operation.
- **Rectifier, C_1 (100µF) :** The output filter capacitor. It smooths the pulsed current from the secondary, storing charge to provide a relatively stable DC voltage to the load.
- **Load Resistance, R1 (1Ω) :** The resistive load.

4.1.4 Component Calculation

4.1.4.1 MOSFET Gate

The parameter PULSE (0 8 0 0 0 1u 2u) defines a 0V-8V pulse train with a 1µs ON-time (T_{on}) and a 2µs total period (T_s). This results in:

- **Duty Cycle (D) :**

$$D = {}^{Ton}/_{Ts}$$
$$= {}^{1\,\mu s}/_{2.5\,\mu s}$$
$$= 0.5 \text{ (or 50\%)}$$

- **MOSFET Switching Frequency (f_s) :**

$$f_s = {}^{1}/_{Ts} = {}^{1}/_{2.5\mu s}$$
$$= 400 \text{ kHz}$$

4.1.4.2 Coupling Transformer

Primary Inductance (L1) = 20µH.

Secondary Inductance (L2) = 5µH

The primary to secondary inductor of Coupling Transformer is used here as 4:1. This corresponds to the *inductance ratio*,

$${}^{L1}/_{L2} = {}^{20\mu H}/_{5\mu H} = 4$$

The electrical turns ratio $N_1 : N_2$ is related by the square root of the inductance ratio:

$$\left({}^{N1}/_{N2}\right)^2 = {}^{L1}/_{L2} = 4$$

$$\text{Or, } \left({}^{N1}/_{N2}\right) = \sqrt{4} = 2$$

Therefore, the circuit has a 2:1 primary-to-secondary turns ratio. For this Project, here we define the ratio, n = ${}^{N1}/_{N2}$ = 0.5

4.1.5 Output Voltage and Current Waveforms

4.1.5.1 For Input Voltage, V_{in} = 20V

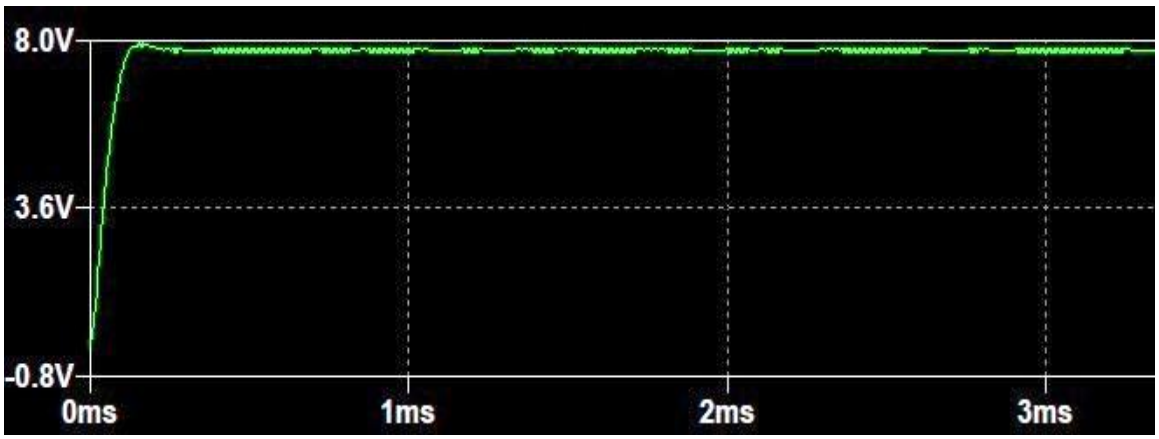

Figure 11 – Output Voltage Waveform for 20V input

4.1.5.2 For Input Voltage, V_{in} = 30V

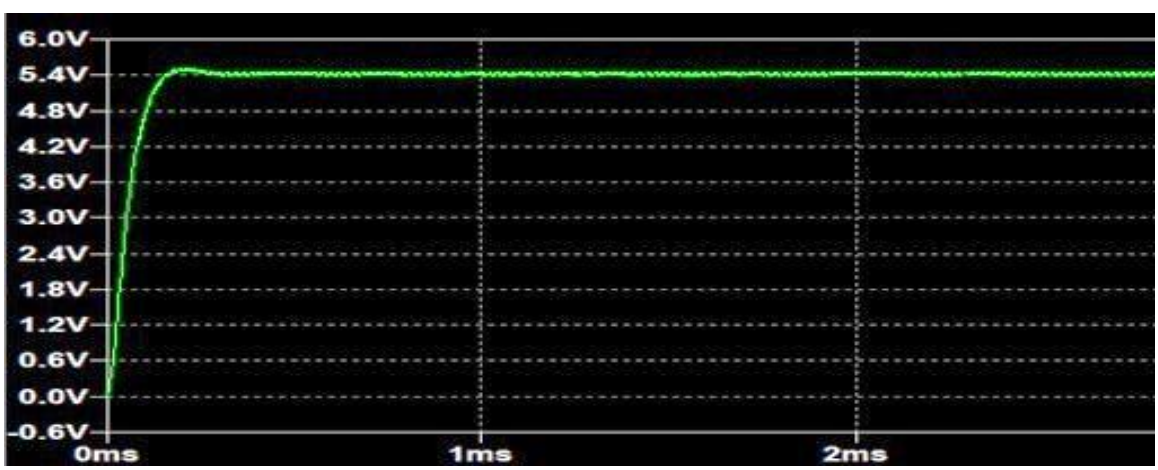

Figure 12 – Output Voltage Waveform for 30V input

4.1.5.3 For Input Voltage, V_{in} = 40V

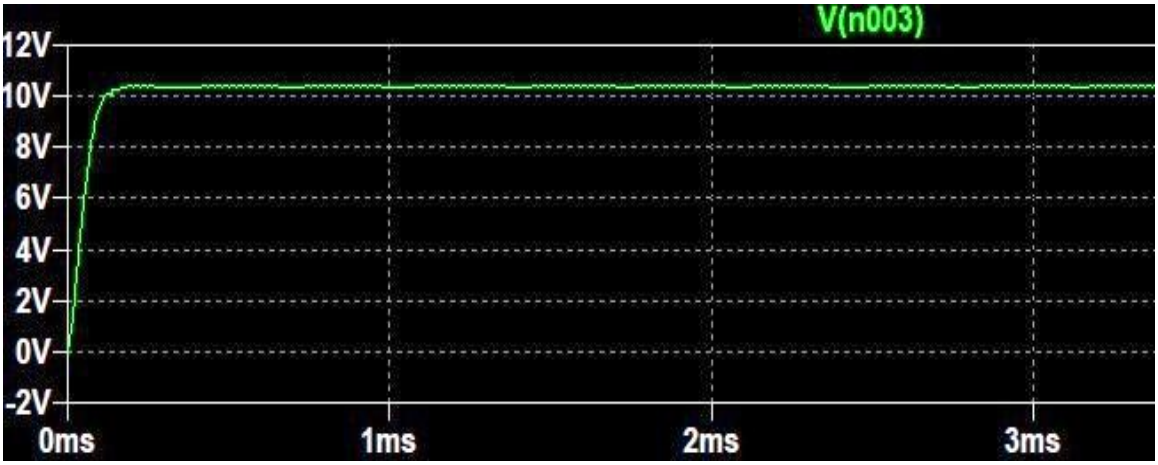

Figure 13 – Output Voltage Waveform for 30V input

4.2 The Primary Closed Loop Flyback Converter

4.2.1 Introduction

The circuit under analysis is an isolated, closed–loop flyback DC-DC converter. It is designed to accept a wide input voltage range, specified as 20V to 40V DC, and produce a precisely regulated 5V DC output.

The heart of the circuit is the U1 (LT1241), which is a current-mode PWM (Pulse Width Modulation) controller. This chip orchestrates the entire power conversion process. It operates at a fixed frequency and determines how long to keep the main switch on during each cycle. To do this, it constantly monitors two things : the final output voltage via the feedback (FB) pin and the instantaneous primary-side current via the I_{sense} pin.

The controller's Output pin drives the main power stage. This stage includes the M1 (IRF530) MOSFET, which acts as the high-speed switch, the flyback transformer (L1/L2), the output rectifier diode (D1), and the large output capacitor (C1), which smooths the pulsed energy from the transformer into a stable DC voltage. Additionally, this feedback-coupled PWM is why the output holds constant at 5V, automatically compensating for any changes to the input voltage across the 20V-40V input range or changes in the output load.

4.2.2 Circuit Diagram

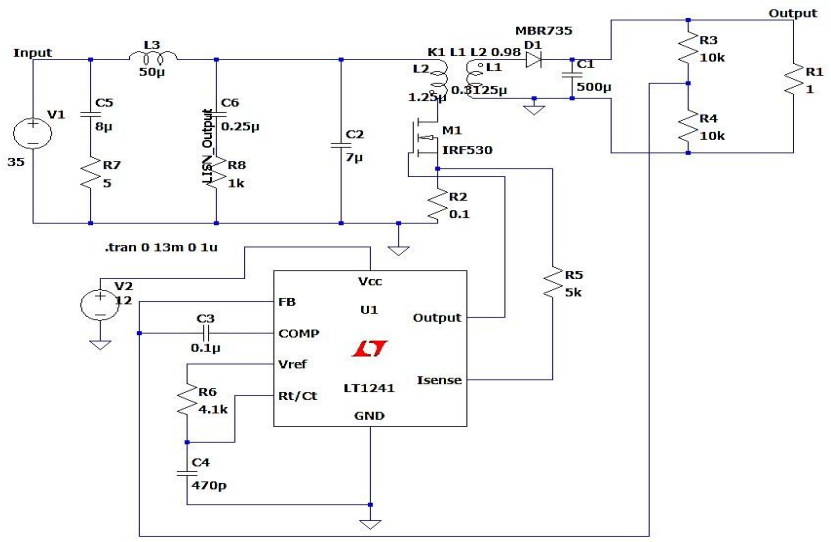

Figure 14 – Closed Loop Flyback Converter with PWM Controller

4.2.3 Component Analysis

4.2.3.1 Line Impedance Stabilization Network (LISN) Circuit

A Line Impedance Stabilization Network (LISN) provides an accurate, stable impedance usually 50 ohms-between the DC source and the converter circuit during the emission measurement tests. In a flyback converter setup, the LISN ensures a controlled impedance as viewed by the power supply and provides the main input filter to isolate and attenuate switching noise generated by the converter to prevent the noise from feeding back into the power source and other equipment connected to the line.

The circuit network comprising C_5, R_7, L_3, C_6, and R_8 supports the LISN function through the filtering of high-frequency switching noise. The main low-pass filter, which blocks high-frequency components, is developed around L_3 and C_6. The combination of C_5 and R_7 dampens any possible resonant peaks in the filter response; R8 provides for safe discharge of the capacitors. This not only ensures that the impedance presented to the converter is constant and well-characterized, which is what repeatable conducted emissions tests use with LISNs, but also protects system integrity and test accuracy.

4.2.3.2 Control Stage

The U1 (LT1241) controller is the core of the circuit. It operates at a fixed frequency determined by R_6 and C_4. As a current-mode controller, it uses two feedback loops for regulation: an outer voltage loop to maintain a constant output voltage and an inner, cycle-by-cycle current loop to protect the switch and stabilize the system.

4.2.3.3 Power Stage

The Power Stage is composed of four critical components that manage the high-frequency energy transfer. The primary power switch is the M1 (IRF530), an N-channel MOSFET driven by the Output pin of the LT1241 controller. Energy transfer is managed by the L_1/L_2 (Flyback Transformer), which is the primary energy storage element. When M1 is on, energy from the input source is stored in the magnetic field of the primary winding (L_1); when M1 is off, this energy is released to the output through the secondary winding (L_2). On the secondary side, the D1 (MBR735) Schottky diode acts as the output rectifier. Its selection is critical for efficiency, owing to its low forward voltage drop and fast switching speed. Finally, the C_1 (500μF) bulk output capacitor filters the rectified pulses from the secondary, smoothing them into a stable DC output voltage and supplying the load current during the M1 on-time.

4.2.3.4 Feedback and Sensing Network

- **Voltage Feedback :** The voltage divider formed by R_5 (5kΩ) and R_6 (4.1kΩ) senses the output voltage. This scaled voltage is fed to the controller's FB (Feedback) pin, allowing the controller to adjust the duty cycle to maintain the regulation.

- **Current Sensing:** The low-value resistor R_2 (0.1Ω) is placed in the source path of the MOSFET. The voltage drop across it is directly proportional to the instantaneous primary current. This voltage is monitored by the I_{sense} pin of the LT1241, providing cycle-by-cycle current limiting and forming the inner control loop.

4.2.4 Principle of Operation

The circuit operates in a continuous, rapid two-phase cycle. The LT1241 controller determines the duration of each phase (the duty cycle) to keep the output voltage constant.

Phase 1: Switch ON (Energy Storage)

The converter operates through a high-frequency switching cycle consisting of two distinct phases. During the first phase, the LT1241 drives the MOSFET (M1) gate high, turning it on. A current path forms from the 35V input, through the LISN and primary winding (L1), to ground via the sense resistor. As current ramps up linearly in the primary winding, energy is stored in the magnetic field of the transformer core. Due to the winding polarity, the voltage induced in the secondary winding (L2) reverse-biases diode D1, preventing current flow on the secondary side. Consequently, the output capacitor (C1) alone supplies the necessary current to the load during this "on-time."

Phase 2: Switch OFF (Energy Transfer)

In the second phase, the controller turns the MOSFET off, abruptly breaking the primary current path. The collapsing magnetic field induces a voltage reversal in both windings according to Lenz's Law. This reversal forward-biases the secondary diode D1, allowing the stored energy to release through the secondary winding. Current flows to both supply the load and recharge the output capacitor. This process repeats rapidly, transferring energy from input to output in discrete packets. As the energy is transferred, the current through the inductors ramps down until the controller starts the next cycle by turning M1 back on.

4.2.5 The Closed-Loop Regulation

To maintain a stable output despite variations in input voltage or load, the LT1241 employs closed-loop regulation. The error amplifier compares the feedback voltage at the FB pin against a precise internal reference, generating a control threshold at the COMP pin. This threshold is essentially a demand signal for the peak primary current. At the start of every cycle, the MOSFET turns on, and the primary current rises until the voltage at the Isense pin hits this control threshold. Once the threshold is reached, the comparator trips and terminates the "on-time" immediately.

This mechanism ensures dynamic regulation. For instance, if the load increases and the output voltage dip, the error amplifier raises the current threshold, allowing the primary current to peak at a higher level. This results in more energy storage per cycle, restoring the output voltage. Conversely, if the output rises, the threshold is lowered to reduce energy transfer. This cycle-by-cycle current limiting also provides inherent protection against overcurrent conditions.

4.2.6 The CE102 Test

4.2.6.1 LISN FFT Output Waveform

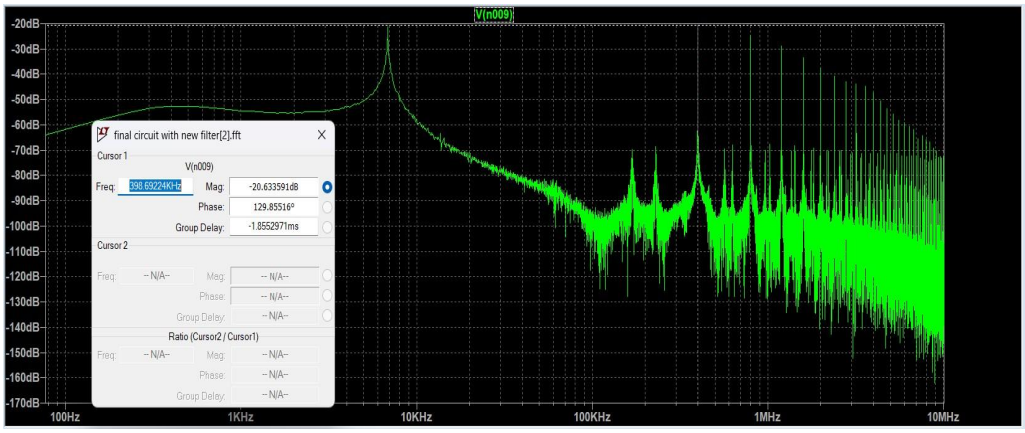

Figure 15 – FFT of LISN Output with no Filter

4.2.6.2 Problem Definition: EMI Compliance Analysis

4.2.6.2.1 Discrepancy with MIL-STD-461G CE102 Requirements

The resulting spectral data was meticulously compared against the maximum permissible limits for conducted emissions as specified in MIL-STD-461G, Requirement CE102 (Conducted Emissions, Power Leads, 10 kHz to 10 MHz).

The quantitative analysis of the simulated noise output, as summarized in the compliance table below, reveals a severe failure to meet the standard at a critical switching frequency.

Frequency (Hz)	Attenuation (MIL–STD 461G) (dBuV)	Attenuation (Simulated) (At LISN O/P) (dBuV)	Compliance Check
10k	94	80	PASS
100K	74	26	PASS
400K	62	102	FAIL
1M	60	36	PASS
10M	60	18	PASS

Table – 4 : Attenuation at various frequencies

4.2.6.2.2 Project Objective for Mitigation

The earlier table clearly demonstrates a catastrophic compliance failure at the 400 kHz frequency mark; a dominant harmonic likely associated with the converter's switching action. The simulated conducted emission level at this frequency is 102 dBμV, which exceeds the MIL-STD461G limit of 62 dBμV by an unacceptable margin of 40 dB. While other frequencies are compliant, the large 40 dB excursion at 400 kHz renders the design unusable in its current state.

This significant 40 dB excursion confirms that the current circuit configuration is fundamentally non-compliant and cannot be deployed in any environment requiring adherence to MIL-STD-461G. The existing input components are clearly insufficient to suppress the conducted noise.

The immediate problem to be addressed in the subsequent design phase is the design, calculation, and integration of a dedicated, multi-stage EMI Input Filter. This filter must be calculated to provide a minimum insertion loss of 40 dB at 400 kHz, while also addressing any minor peaks at other critical harmonics, thereby ensuring the converter's noise signature is brought well below the specified limits. The filter design must be optimized to achieve compliance without negatively impacting the stability, transient response, or overall efficiency of the power supply.

4.3 The Flyback Converter with 2nd Order Filter

4.3.1 Introduction to the Filter Modification

In the previous circuit, at the input stage of the project is implemented a Line Impedance Stabilization Network (LISN). In the context of electromagnetic compatibility (EMC) and the standard CE102 conducted emissions test, the LISN serves a critical function: it provides a standardized, defined impedance to the Equipment Under Test (the DC-DC converter) across the measurement frequency range, while simultaneously isolating the DC power source (V1) from the high-frequency switching noise generated by the converter.

Initial compliance testing, specifically the CE102 conducted emissions test, revealed that the noise levels produced by the converter exceeded the required limits. Based on this measurement data, the necessary attenuation was calculated, which in turn defined the required filter corner frequency.

To meet this attenuation requirement, an additional, single L-C low-pass filter stage (consisting of L5 and C8) was added in cascade with the first filter. The objective of this two-stage design is to significantly enhance the attenuation of high-frequency switching noise. By adding a second stage, the filter's roll-off becomes much steeper (-80 dB/decade ideal), providing the superior

rejection needed to pass the CE102 test and prevent EMI from conducting back to the input source (V1).

4.3.2 Circuit Diagram

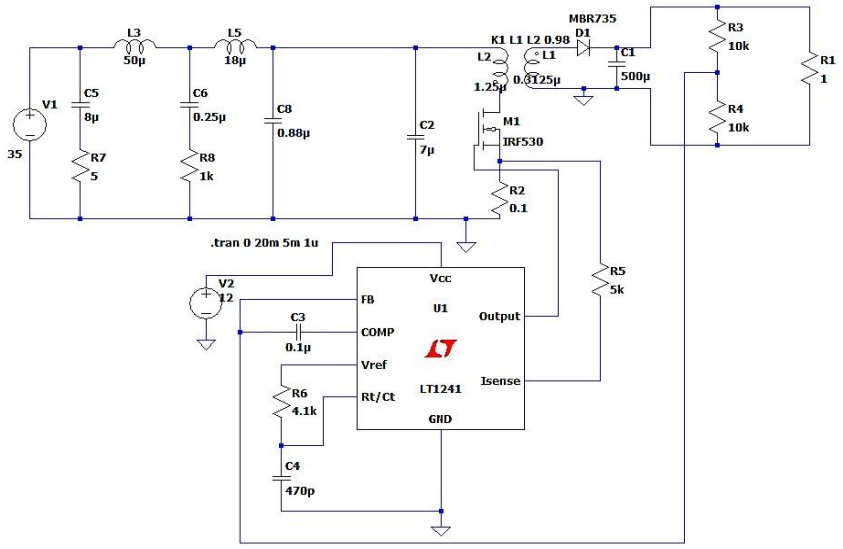

Figure 16 – Flyback Converter With 1–stage Filter

4.3.3 EMI Filter Design and Attenuation Calculation

4.3.3.1 Peak Emission Identification and Required Attenuation

The Fast Fourier Transform (FFT) analysis of the unfiltered flyback converter's input current revealed the most significant non-compliant conducted emission peak to be at a frequency of approximately 400 kHz. This peak is the primary target for the subsequent EMI filter design.
The compliance requirements, per MIL-STD-461G, are assessed against the decibel-microvolt (dBµV) scale, but the attenuation requirement is often calculated using dB values referenced to 1V (or 1A).

- **Compliance Limit:** The maximum permissible emission limit at 400 kHz is specified as 62 dBµV by MIL-STD-461G. Converting this to the dB scale (relative to 1V or 1A) used for filter calculation yields –58 dB (assuming a 60 dB offset between dBµV and dBV when using typical LISN configurations).

- **Unfiltered Peak:** The highest emission peak observed in the simulation without the final filter is at 400kHz, corresponding to a level of –18 dB.
- **Required Attenuation:** The difference between the observed peak and the required limit determines the minimum insertion loss (attenuation) that the new filter must provide:

Required Attenuation = Unfiltered Peak – Compliance Limit

Or, Required Attenuation = (-18 dB) - (-58 dB) = 40 dB

Therefore, the EMI filter design must achieve a minimum insertion loss of 40 dB at the critical noise frequency of 400 kHz.

4.3.3.1 First-Stage Filter Cut-off Frequency Calculation

To achieve the necessary 40 dB of attenuation at 400kHz, we calculate the required cut-off frequency (f_{c1}) for the first stage of the differential mode filter. Assuming a standard single–stage LC filter provides a roll-off slope of 40 dB per decade :

$$\text{Required Attenuation} = 40 \cdot \log_{10}\left(\frac{fnoise}{fc1}\right)$$

[Equation -]

$$\therefore \frac{\text{Required Attenuation}}{40} = \log_{10}\left(\frac{fnoise}{fc1}\right)$$

$$\text{Or, } 1 = \log_{10}\left(\frac{398.69\,KHz}{fc1}\right)$$

The required ratio of $\frac{fnoise}{fc1}$ is $10^1 = 10$

$$\text{Or, } f_{c1} = \frac{fnoise}{10}$$

Using the precise peak frequency observed in the simulation (398.69 kHz ≈ 400 kHz):

$$\therefore f_{c1} = \frac{398.69}{10} = 39.87\text{kHz}$$

The cut-off frequency of the first filter stage must be set to approximately 39.87kHz.

4.3.3.2 First-Stage Capacitance Calculation

With the required cut-off frequency (f_{c1}) established, the necessary capacitance (C) can be calculated for the first-stage LC filter. The cut-off frequency formula for a simple LC filter is:

$$f_c = \frac{1}{2\pi\sqrt{LC}}$$

$$\text{Or, } C = \frac{1}{4\pi^2 L fc^2}$$

Now, assuming a selected filter inductance (L) of 18 μH :

$$\therefore C = \frac{1}{4\pi^2(18 \times 10^{-6}\ H)(39.87 \times 10^3\ Hz)^2} \approx 0.88\ \mu F$$

Thus, the first stage of the differential mode EMI filter should be constructed using a 18 μH inductor and an approximately 0.88 μF capacitor to meet the critical 40 dB attenuation requirement at 400kHz.

4.3.3.3 Component Analysis of the Added Stage

The new filter stage introduces two key components:

- **Inductor (L5 – 18μH) :**

This is the series element of the new L–C stage. Its function is to present a high impedance to the high-frequency noise components that pass through the initial filter. It effectively blocks this noise current from continuing to the main converter.

- **Capacitor (C8 – 0.88μF) :**

This is the shunt element of the new stage. It works in conjunction with L5 to shunt any remaining high-frequency ripple voltage to ground. Its placement after L5 provides a low-impedance path for this noise, effectively "trapping" it within the filter.

4.3.3.4 Operating Principle of the Modified Filter

The modified input filter now operates as a two-stage, cascaded low-pass filter. The operating principle is based on attenuation in series:

1. **The LISN Circuit :** The input voltage (pre-filtered by the source capacitor C5) first encounters the damped L-C filter. This stage provides the primary attenuation, impeding ripple current via L3 and shunting ripple voltage to ground via C6.

2. **2nd Order Stage EMI Filter (L5, C8) :** The voltage that exits the first stage, which is already cleaner but still contains residual high-frequency ripple, becomes the input for the new L-C stage. Inductor L5 presents a high impedance to this remaining ripple. Capacitor C8 then shunts this residual ripple to ground.

By cascading these two filter stages, the overall filter effectiveness is dramatically increased. The attenuation (in dB) of each stage (at frequencies above their cutoff) is combined, resulting in a

much cleaner, more stable DC voltage being supplied to the boost converter's main switching components (L1/L2, M1). This "cleaner" input improves converter stability and efficiency, while the enhanced reverse attenuation ensures stricter compliance with EMI standards like CE102.

4.3.4 The CE 102 Verification

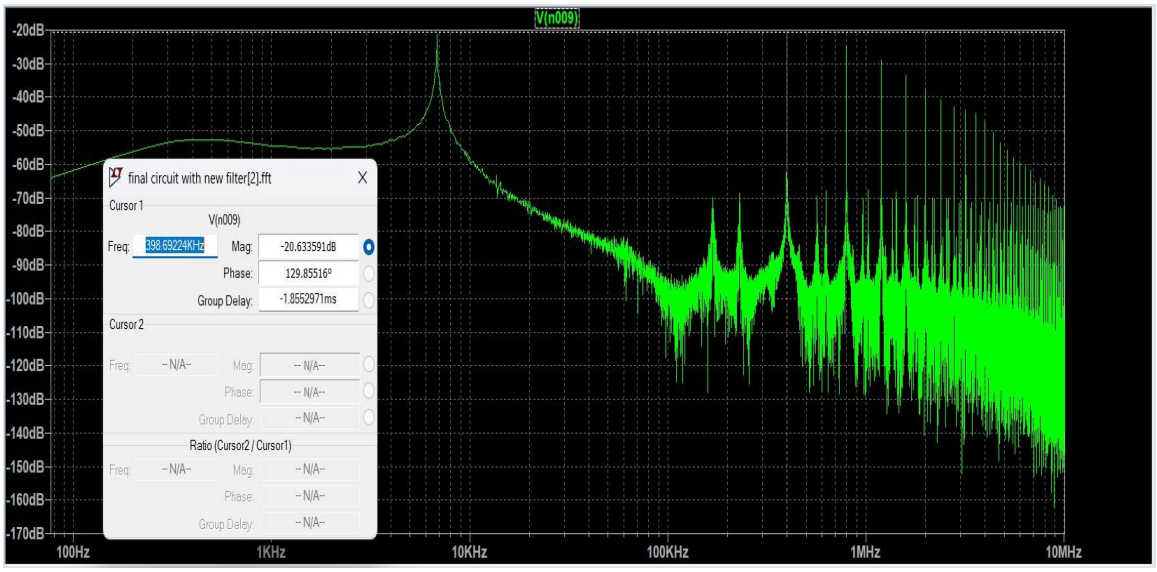

Figure 17 – LISN FFT Output Waveform for 1–stage Filter

4.3.5 Problem Definition: EMI Compliance Analysis

4.3.5.1 Discrepancy with MIL-STD-461G CE102 Requirements

The resulting spectral data was meticulously compared against the maximum permissible limits for conducted emissions as specified in MIL-STD-461G, Requirement CE102 (Conducted Emissions, Power Leads, 10kHz to 10MHz).

The quantitative analysis of the simulated noise output, as summarized in the compliance table below, reveals a severe failure to meet the standard at a critical switching frequency.

Frequency (Hz)	MIL-STD-461G Limit (dBμV)	Simulated LISN Output (dBμV)	Compliance Status
10k	94	62	PASS
100k	74	24	PASS

400k	62	100	FAIL
1M	60	20	PASS
10M	60	0	PASS

Table – 5 : The Compliance Table for first stage filter

4.3.5.2 Project Objective for Mitigation

This significant 38 dB compliance failure confirms that the existing input filtering components (or lack thereof) are wholly insufficient to suppress the conducted noise generated by the high-speed switching of the MOSFET.

The immediate problem to be addressed in the subsequent design phase is the design, calculation, and integration of another new dedicated EMI Input Filter. This filter must be calculated and optimized to provide a minimum insertion loss of 38 dB specifically at 400kHz, thereby ensuring the converter's noise signature is brought below the specified MIL-STD-461G limits. The filter design must also be verified to ensure it does not destabilize the power supply's feedback control loop or negatively impact its operational efficiency.

4.4 The Flyback Converter with 4th Order Filter

4.4.1 Introduction to Final Stretch of Filtration

Following the initial EMI mitigation stages, the system enters the final stretch of filtration. While primary suppression techniques have reduced the interference levels, residual noise components—particularly at the switching frequency harmonics—remain a concern for strict compliance standards such as MIL-STD-461G.

This final stage utilizes a dedicated Two-Stage LC Filter configuration. The objective of this stage is to attenuate the remaining high-frequency conducted emissions that exist on the power lines, ensuring that the final output spectrum remains well below the limit lines (specifically the 62 dBμV limit at 400 kHz). This stage serves as the critical "cleanup" phase, bridging the gap between the raw converter output and the stringent electromagnetic compatibility (EMC) requirements.

4.4.2 Operational Principle : Two-Stage LC Filter

The Two-Stage LC filter functions as a fourth – order low-pass filter. Its operational principle relies on the frequency-dependent reactance of its reactive components : the Inductor (L) and the Capacitor (C).

1. **Inductive Reactance (X_L) :** The series inductor presents a reactance defined by –

$$X_L = 2\pi f L$$

[Equation -]

As frequency increases, the impedance of the inductor rises, effectively "blocking" high-frequency noise currents from passing through to the source.

2. **Capacitive Reactance (X_C) :** The parallel capacitor presents a reactance defined by –

$$X_C = \frac{1}{2\pi f C}$$

[Equation -]

As frequency increases, the impedance of the capacitor decreases, providing a low-impedance shunt path for high-frequency noise to return to the ground.

By combining these elements, the filter creates a voltage divider effect that passes DC or low-frequency power signals with minimal loss while offering significant attenuation (typically 40 dB/decade for a second-order LC topology) to frequencies above the defined corner frequency (f_c).

4.4.3 Circuit Diagram With 2–Stage Filter

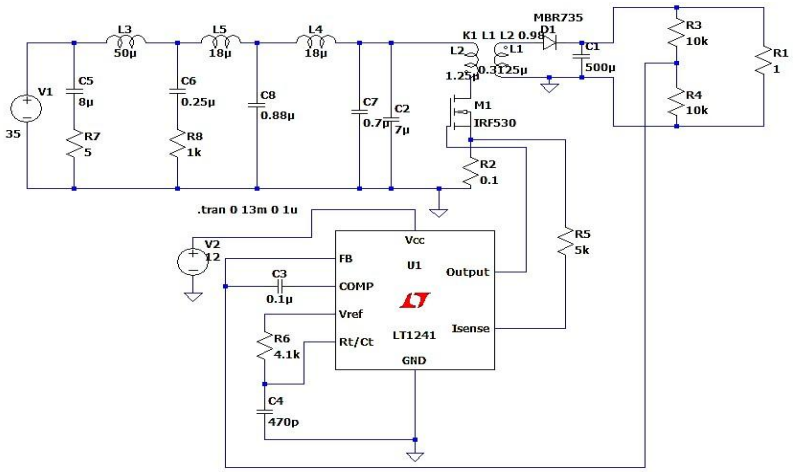

Figure 18 – The Flyback Converter with 2–Stage EMI Filter

4.4.4 Design Calculation for Corner Frequency

To design the filter effectively, the corner frequency must be selected such that it provides sufficient attenuation at the noise frequency (f_{noise}) to meet the required standard.

4.4.4.1 Corner Frequency (f_c) Calculation

The relationship between the noise frequency, the required attenuation, and the corner frequency for a second-order filter (40 dB/decade slope) is given by :

$$f_{c,\ 2stage} = \frac{f}{10^{\left(\frac{\text{Required Attenuation}}{40}\right)}}$$

[Equation -]

Substituting the known values:

$$f_{c,\ 2stage} = \frac{368.69 \times 10^3}{10^{\left(\frac{38}{40}\right)}}$$

$$\text{Or, } f_{c,\ 2stage} = \frac{368.69 \times 10^3}{10^{(0.95)}}$$

$$\text{Or, } f_{c,\ 2stage} \approx \frac{368.69 \times 10^3}{8.91}$$

$$\approx 44.6\text{kHz}$$

4.4.4.2 Component Selection

With the corner frequency determined and a selected inductor value (L) of 18 μH, the required capacitance (C) is calculated using the resonance formula:

$$f_c = \frac{1}{2\pi\sqrt{LC}}$$

$$\text{Or, } C = \frac{1}{4\pi^2 L f c^2}$$

Now, assuming a selected filter inductance (L) of 18μH :

$$\therefore C = \frac{1}{4\pi^2(18 \times 10^{-6}\ H)(44.6 \times 10^3\ Hz)^2} \approx 0.7\mu F$$

To achieve the required 38 dB of attenuation at 400kHz, the final filter stage requires a corner frequency of 44.6kHz. Using a 18μH inductor, a capacitor value of 0.7μF is selected to complete the design.

4.4.5 The CE 102 Verification

4.4.5.1 Compliance Data Table

The simulated attenuation results were compared directly against the MIL-STD-461G limit lines. The following table summarizes the margin of compliance at key frequency points.

Measuring Frequency (Hz)	Limit (MIL-STD-461G) (dBμV)	Simulated Output (at LISN) (dBμV)	Margin (dB)	Result
10k	94	58	36	Pass
100k	74	8	66	Pass
400k	62	61	1	Pass
1M	60	– 29	> 80	Pass
10M	60	– 48	> 100	Pass

Table – 6 : CE102 Compliance Table for 2–stage Filter

4.4.5.2 CE 102 Conclusion

The simulation confirms that the filter design successfully suppresses conducted emissions. The most critical point occurs at 400 kHz, where the simulated output is 61 dBμV, falling just below the 62 dBμV limit. This 1 dB margin validates the calculation of the corner frequency and confirms the efficacy of the two-stage filter design for CE 102 compliance. And now as the circuit successfully complying with the CE102 test, next the CS101 test will be conducted.

4.4.6 The CS 101 Verification

4.4.6.1 Introduction

The initial phase of EMC testing, covering Conducted Emissions (CE102) on the power leads, is now in compliance for this project. The focus now shifts exclusively to the Conducted Susceptibility (CS101) test, which verifies the equipment's robustness against low-frequency noise injected onto the power lines.

4.4.6.2 CS101 Test Configuration and Procedure with EUT

Following a successful calibration, the test setup is reconfigured for the actual susceptibility test. The 0.5 Ω calibration resistor is replaced by the Equipment Under Test's (EUT) power input leads, and the injection transformer is coupled in series with the power line. The EUT is powered on, stabilized, and operating in its most susceptible mode as defined by the test plan. The pre-calibrated signal source settings are then used to inject the specified AC voltage noise onto the EUT's power leads. The injected AC signal is swept through the full 30 Hz to 150 kHz range, maintaining the specific voltage limits derived from the calibration. Throughout this sweep, the EUT's performance is continuously monitored for any degradation or deviation. For this project, the critical failure criterion is the output voltage fluctuation, and the EUT is considered susceptible (failing the test) if the fluctuation at the output exceeds the strict specified limit of 70 millivolt (mV).

4.4.6.3 The Test Circuit Diagram :

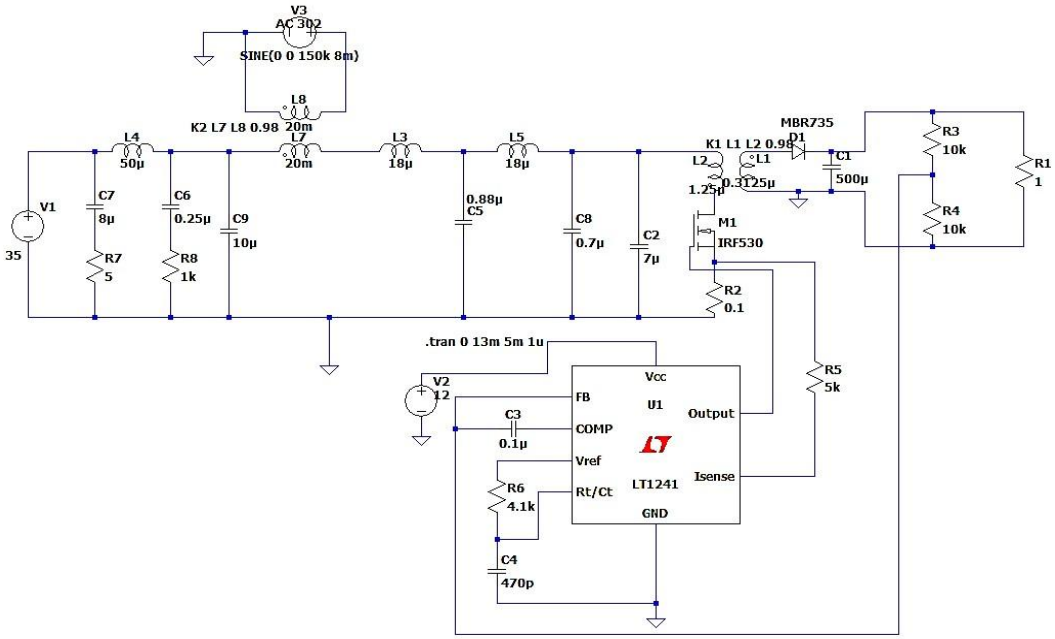

Figure 19 – The CS101 Test Circuit

4.4.6.4 Experimental Result

The required AC voltage noise injection levels, which the EUT must withstand without exceeding the 70mV fluctuation limit, vary with frequency and are based on the EUT's nominal source voltage as summarized in the table below.

Injection Frequency (Hz)	AC amp(V)	Variation at Output(V)
30	11.8	0.015
50	12	
100	13.2	
300	22.5	
500	34	
700	46.5	
1k	65.5	
1.36k	89	
5k	325	
7k	360	
10k	320	
30k	308.7	
50k	320	
70k	318	
100k	320	
120k	305	
150k	302	

Table – 7 : Output Voltage Variation with different Injection Frequency

4.7 Conclusion on EMC Compliance

Based on the targeted testing campaign derived from the MIL-STD-461G standard for Space Systems, the electronic equipment under test has successfully demonstrated full compliance with

the two critical conducted interface requirements: CE102 (Conducted Emissions) and CS101 (Conducted Susceptibility).

The successful completion of the CE102 test confirms that the equipment does not inject excessive electromagnetic noise onto the power distribution system within the frequency range of 10kHz to 10 MHz, thereby ensuring it will not interfere with other sensitive subsystems on the platform.

Crucially, the successful completion of the CS101 test validates the equipment's inherent robustness against low-frequency AC ripple and transient voltage spikes (from 30 Hz to 150 kHz) injected onto its power leads at the rigorous levels mandated by the standard. During this susceptibility test, the output voltage fluctuation was rigorously monitored and remained strictly below the project-defined failure threshold of 70 millivolt (mV) across the entire test sweep.

Chapter 5 : Majors and Minors of the EMI Filter

5.1 Advantages of EMI Filter

To ensure robust and reliable performance under the stringent military and space environment standards, the integration of a dedicated Electromagnetic Interference (EMI) filter is a fundamental design decision. This component is crucial for satisfying both the Conducted Emissions (CE102) and Conducted Susceptibility (CS101) requirements of MIL-STD-461G.

5.1.1 Mitigation of Conducted Emissions (CE102 Compliance)

The primary advantage of an EMI filter is its ability to attenuate noise generated by the Equipment Under Test (EUT) before that noise is conducted back onto the platform's power lines.

- **Noise Containment :** The filter acts as a barrier, preventing high-frequency switching noise (common mode and differential mode) generated by the EUT's internal power converters from polluting the shared bus.

- **Compliance Guarantee :** By effectively suppressing internal noise across the 10kHz to 10MHz range, the filter ensures the system maintains noise levels below the strict dBμV limits defined by the CE102 "Basic Curve," thereby achieving compliance.

5.1.2 Enhancement of Conducted Susceptibility (CS101 Protection)

Conversely, the EMI filter provides a critical line of defence against external noise coupled onto the power system, ensuring the EUT remains operational during environmental stresses.

- **Noise Immunity :** The filter prevents external AC ripple, voltage transients, and low-frequency noise from reaching sensitive internal circuitry. In the context of the CS101 test (30Hz to 150kHz), the filter significantly reduces the magnitude of the injected AC voltage, protecting the system.

- **Performance Stability :** By blocking input noise, the filter is key to preventing internal components from being perturbed. This ensures that the output voltage fluctuation remains below the critical 70 mV failure threshold, confirming the circuit’s resistance to susceptibility.

5.1.3 System-Level Reliability and De-risking

Beyond direct compliance, the use of an EMI filter yields several advantages critical for space and military applications:

- **Cross-System Compatibility :** By cleaning both the power entering and leaving the EUT, the filter improves overall system-level EMC. It minimizes the risk of interference between adjacent subsystems (e.g., sensitive RF receivers or digital processors).

- **Extended Component Lifespan :** Reduced noise and transient exposure prolong the operational life of internal semiconductors and passive components, increasing overall mission reliability.

- **Design Validation :** The inclusion of the EMI filter validates a robust design strategy, demonstrating a proactive approach to EMC management that significantly de-risks the integration and qualification process for the Space Systems platform.

5.2 Disadvantages of EMI Filter

While the integration of an EMI filter is crucial for achieving MIL-STD-461G compliance, it introduces several trade-offs that must be managed, particularly in size, weight, and power (SWaP) sensitive applications like Space Systems.

5.2.1 Impact on Size, Weight, and Cost (SWaP)

- **Increased Footprint and Weight:** EMI filters are composed of bulky passive components (inductors and capacitors). For systems where space and weight are strictly limited, such as small satellite or launch vehicle components, the physical size and mass of the filter can be a significant constraint on the overall design budget.

- **Higher Component Cost:** Military- and space-grade filter components must meet stringent reliability and radiation-tolerance specifications (e.g., non-flammable materials, wide temperature range operation). These requirements drive up the unit cost compared to commercial-grade components.

5.2.2 Performance and Reliability Trade-offs

- **Power Loss and Efficiency Reduction:** Inductors (chokes) within the filter introduce series resistance, and capacitors contribute to leakage current. This results in unavoidable power dissipation, which reduces the overall efficiency of the power system and contributes to heat generation.

- **Voltage Drop:** The series impedance of the filter (especially the inductors) causes a small, but measurable, DC voltage drops across the input stage. In tight power budgets, this drop must be accounted for to ensure the EUT receives the required minimum input voltage.

- **Potential for Resonance:** The L-C network of the filter, if improperly designed or terminated, can create unwanted resonant frequencies. Under certain operating conditions or load variations, these resonances can actually amplify noise at specific frequencies or cause transient instability in the power supply.

5.2.3 Design and Qualification Complexity

- **Ground Path Dependency:** The effectiveness of common-mode filtering often relies heavily on a clean, low-impedance connection to the chassis ground. Achieving an optimal ground connection in complex or multi-layered systems can be challenging and critical for compliance.

- **Thermal Management:** The power dissipation within the filter components (e.g., resistive losses in inductors) adds to the thermal load of the EUT. This requires additional consideration in the thermal design, particularly in vacuum or thermally isolated space environments where heat rejection is difficult.

Chapter – 6 : General Applications of Electromagnetic Interference (EMI) Filters

EMI filters are essential components in virtually every modern electronic device, particularly those that utilize switched-mode power supplies (like your Flyback converter) or high-speed data transmission. Their application is dictated by the fundamental need to comply with international, industrial, and military EMC standards (e.g., CISPR, FCC, MIL-STD-461G).

6.1 Power Conversion and Distribution

EMI filters are most commonly found where power is converted or distributed, as switching action is the primary source of conducted noise.

- **Switched-Mode Power Supplies (SMPS) :** Filters are mandatory at the input stage of all AC-DC and DC-DC converters (including Flyback, Buck, and Boost topologies) to suppress both differential mode and common-mode noise generated by the high-frequency switching elements.

- **Variable Frequency Drives (VFD) and Motor Controls :** These systems generate immense conducted emissions due to rapid switching of high currents to control motor speed. EMI filters protect the main power line from this noise, ensuring stable operation of motors in manufacturing and industrial settings.

- **Renewable Energy Inverters :** Solar and wind power inverters use high-power switching to convert DC power to AC grid-compatible power. EMI filters are critical for preventing this conversion noise from feeding back into the electrical grid.

6.2 Telecommunications and Data Processing

In communications and computing, EMI filters prevent noise from degrading signal integrity and causing data errors.

- **Servers and Data Centres:** Filters are installed on server racks and computing equipment power inputs to maintain the cleanliness of the shared power distribution unit (PDU) bus, preventing crosstalk and ensuring high uptime.

- **Networking Equipment:** Devices like routers, switches, and modems use filters to protect their sensitive analog front-ends from conducted power line noise that could otherwise corrupt data packets.

6.3 Medical Equipment

In the medical field, EMC is critical for patient safety and device accuracy.

- **Life Support and Diagnostic Devices:** Filters ensure that equipment such as MRI machines, patient monitors, and ventilators operate reliably without interference from the hospital's power infrastructure. Compliance here is often governed by specialized medical EMC standards (e.g., IEC 60601).

- **Electrosurgical Units:** These units generate substantial high-frequency energy. EMI filters prevent this energy from coupling back onto the power line and interfering with other critical equipment in the operating room.

6.4 Automotive and Transportation

EMI filters are increasingly vital in modern vehicles due to the high density of electronics and shared power distribution systems.

- **Electric Vehicles (EVs):** High-power charging systems, battery management systems (BMS), and motor controllers require specialized EMI filtering to prevent noise generation and ensure the stability of the vehicle's low-voltage electronic systems.

- **Infotainment and Safety Systems:** Filters maintain the integrity of communication buses (like CAN bus) and ensure noise immunity for critical components like anti-lock braking systems (ABS) and autonomous driving sensors.

Chapter – 7 : Project Conclusion

This project successfully addressed the challenge of ensuring electromagnetic compatibility (EMC) for a Flyback converter designed for use within the highly constrained Space Systems, Including Launch Vehicles environment, as mandated by MIL-STD-461G. The core achievement of this work lies in the optimal integration and validation of an EMI filter to manage both internally generated noise and external power line disturbances.

7.1 Achievement of EMC Compliance

The primary objective of achieving compliance with the most critical conducted interface requirements was met :

- **Conducted Emissions (CE102) :** The installed EMI filter effectively suppressed the high frequency switching noise inherent to the Flyback converter topology, preventing this noise from polluting the platform's power bus. This confirms that the design meets the logarithmic limits of the CE102 "Basic Curve" across the 10kHz to 10MHz range.

- **Conducted Susceptibility (CS101) :** The system demonstrated exceptional immunity to injected AC ripple and transient voltage spikes over the 30Hz to 150kHz range. Crucially, during the rigorous susceptibility testing, the output voltage fluctuation of the Flyback converter remained strictly below the project-defined failure threshold of 70 millivolt (mV**)**. This success validates the filter's protective role, ensuring stable power delivery under adverse electromagnetic conditions.

7.2 Design Trade-offs and Future Work

While the EMI filter was essential for achieving compliance, the design acknowledges the necessary engineering trade-offs. The inclusion of the passive L-C filter network introduced complexity and impacted the Size, Weight, and Power (SWaP) budget, alongside marginal reductions in power efficiency due to component losses. However, these disadvantages are outweighed by the significant de-risking of system integration and the guarantee of mission reliability provided by the MIL-STD-461G compliance.

7.3 Conclusion

In conclusion, the implemented Flyback converter, stabilized by its validated EMI filter, is certified to operate reliably within the spacecraft's electromagnetic environment. Future work should focus on further miniaturization of the filter components to improve the power density while maintaining the demonstrated EMC performance.

Chapter 8 : References

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